



Higher Diploma in Science in Computing

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B8IT131 PROJECT

Container Management System for Reusable Production Containers

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1. Introduction

Reusable production containers play an important role in pharmaceutical manufacturing and must be managed to maintain traceability and compliance with Good Manufacturing Practice (GMP). Before entering production, containers must be cleaned and approved by Quality Assurance (QA). They may be used for a maximum of fourteen production runs or thirty calendar days before requiring cleaning and Supervisor approval to begin a new lifecycle.

Many organisations still rely on paper records to manage these activities, increasing the risk of human error, poor traceability, and compliance issues. This project addresses these challenges by developing a full-stack Container Management System that digitises the container lifecycle, manages approvals, enforces lifecycle rules, and provides secure user authentication, audit logging, and operational dashboards.

At the time of writing, the core functionality has been implemented, including lifecycle management, approval workflows, movement tracking, QR code and barcode functionality, and a simulation engine. The remaining work focuses on system testing, user interface refinements, operational data evaluation, and completion of the final dissertation.

2. Background

Manufacturers working in regulated industries such as pharmaceutical manufacturing must maintain accurate records and provide complete traceability throughout the production process. Many companies still rely on paper records, spreadsheets, and manual approvals to manage reusable production containers. While these methods may work for smaller operations, they become more difficult to manage as production increases and are more likely to result in missing information, human error, and poor traceability.

Good Manufacturing Practice (GMP) requires organisations to demonstrate that reusable containers have been cleaned, approved, and used within their approved lifecycle. This project addresses these challenges by developing a digital Container Management System that records container movements, manages approvals, enforces lifecycle rules, and provides a complete audit trail. The system is intended for production operators, Quality Assurance (QA) staff, supervisors, and administrators, and has been developed using an Agile Scrum approach to support incremental development and continuous testing.

3. Project Scope and Objectives

The aim of this project is to design and develop a full-stack Container Management System to manage the complete lifecycle of reusable production containers used in pharmaceutical manufacturing. The system replaces manual paper-based processes with a digital solution that improves traceability, supports regulatory compliance, and provides greater visibility of container activity.

The scope of the project includes secure user authentication, role-based access control, container lifecycle management, workflow validation, movement tracking, QA and Supervisor approval workflows, QR code and barcode functionality, operational dashboards, and audit logging. A simulation engine has also been developed to generate realistic operational data for testing and evaluation.

The objectives of the project are to:

- Design and implement a relational MySQL database to store container, user, movement, and approval data.
- Develop a RESTful API using Node.js and Express to support the application.
- Create a responsive user interface using React and Material UI.
- Implement business rules to enforce the container lifecycle and approval workflow.
- Provide complete traceability through movement history and audit logging.
- Develop dashboards to monitor container activity and operational performance.
- Generate simulated operational data to support testing and system evaluation.

4. Student's Learning objectives

The primary learning objective of this project was to develop practical software engineering skills by designing and implementing a real-world application for managing reusable production containers. The project has provided experience across the Software Development Life Cycle (SDLC), including system design, full-stack development, testing, and documentation. It has also strengthened knowledge of React, Node.js, Express, MySQL, and REST API development while highlighting the importance of workflow automation, traceability, and regulatory compliance.

Developing the application required integrating multiple technologies, implementing business rules, designing a relational database, and creating a responsive user interface. Overcoming technical challenges throughout the project reinforced the importance of problem solving, testing, and iterative software development.

The learning objectives for this project are to:

- Strengthen full-stack development skills using React, Node.js, Express, and MySQL.
- Apply software engineering principles throughout.
- Develop a secure RESTful web application using authentication, role-based access control.
- Design and implement a relational database.
- Gain practical experience implementing business rules and workflow validation within a real-world software application.
- Develop interactive dashboards and data visualisations using Chart.js to present operational information and key performance indicators.
- Generate realistic operational datasets through simulation to support system testing, evaluation, and analytical reporting.
- Gain practical experience using Git, GitHub, Thunder Client, Jest.
- Develop project management skills through the application of Agile Scrum principles.
- Improve problem-solving, debugging, and software testing skills by resolving technical challenges encountered during the implementation of the project.
- Develop a greater understanding of how digital workflow management systems can improve operational efficiency, traceability, and compliance within pharmaceutical manufacturing.

5. Technical Specification and Design

The Container Management System uses a full-stack architecture that separates the frontend, backend, and database, making the application easier to maintain and extend. The React frontend communicates with a RESTful API developed using Node.js and Express, while a MySQL database stores application data.

The frontend was developed using React, Vite, Material UI, Axios, and JSON Web Tokens (JWT) to provide a secure and responsive user interface. The backend manages authentication, business rules, approval workflows, and container lifecycle validation through REST API endpoints.

A simulation engine generates realistic operational data for testing, dashboards, and reporting. The project was developed and tested using the following hardware and software technologies.

Hardware

- Windows-based personal computer
- Internet connection

Software and Development Tools

- Visual Studio Code – Integrated Development Environment (IDE)
- Node.js – JavaScript runtime environment
- Express – Backend web application framework
- React with Vite – Frontend development framework
- Material UI – User interface component library
- MySQL – Relational database management system
- MySQL Workbench – Database administration and design
- Git – Version control
- GitHub – Source code repository
- Thunder Client – REST API development and testing
- Axios – Client-server communication
- JSON Web Tokens (JWT) – Authentication and authorisation
- Chart.js – Dashboard visualisation and reporting
- Jest – Backend unit testing
- npm – Package management
- Trello – Agile Scrum project planning and sprint management
- Figma – User interface wireframing
- Draw.io – UML and workflow diagrams
- Lucid chart – System and database diagrams

Database

A MySQL relational database stores users, containers, movements, approvals, and audit logs. Primary and foreign keys maintain data integrity and support workflow validation and traceability. **Testing**

Testing has been carried out throughout development using Thunder Client, Jest, and MySQL queries. Authentication, container lifecycle rules, approval workflows, movement tracking, dashboards, and data integrity have all been successfully tested.

6. Project Testing and Evaluation

Testing has been carried out throughout development to verify that the application functions correctly and enforces the container lifecycle rules. Thunder Client has been used to test REST API endpoints, while the frontend has been tested to ensure users can manage containers, approvals, movement history, and dashboards.

Business rules, including lifecycle limits and QA and Supervisor approval workflows, have been tested to confirm that containers cannot progress unless all required conditions are met. The simulation engine has also been used to generate operational data for testing dashboards and reporting.

At the time of writing, the core functionality has been successfully tested. The remaining work will focus on end-to-end testing, user interface refinements, and user acceptance testing before the final submission.

7. Demonstration of Progress

Significant progress has been made in developing the Container Management System. The core functionality of the application has been implemented and is operational. Users can securely log into the system, manage reusable production containers, record container movements, and complete QA and Supervisor approval workflows. The application also enforces lifecycle rules to ensure containers cannot exceed their approved usage limits.

Operational dashboards have been developed to provide an overview of container status and activity. QR code and barcode functionality has also been implemented, allowing containers to be quickly identified and retrieved. In addition, a simulation engine has been developed to generate realistic operational data for testing and evaluating the system.

Figures included in this report demonstrate the current state of the application, including the login screen, dashboard, container management interface, approval workflow, QR code scanner, and analytics dashboard. These screenshots show that the core objectives of the project have been achieved and provide evidence of the progress made to date.

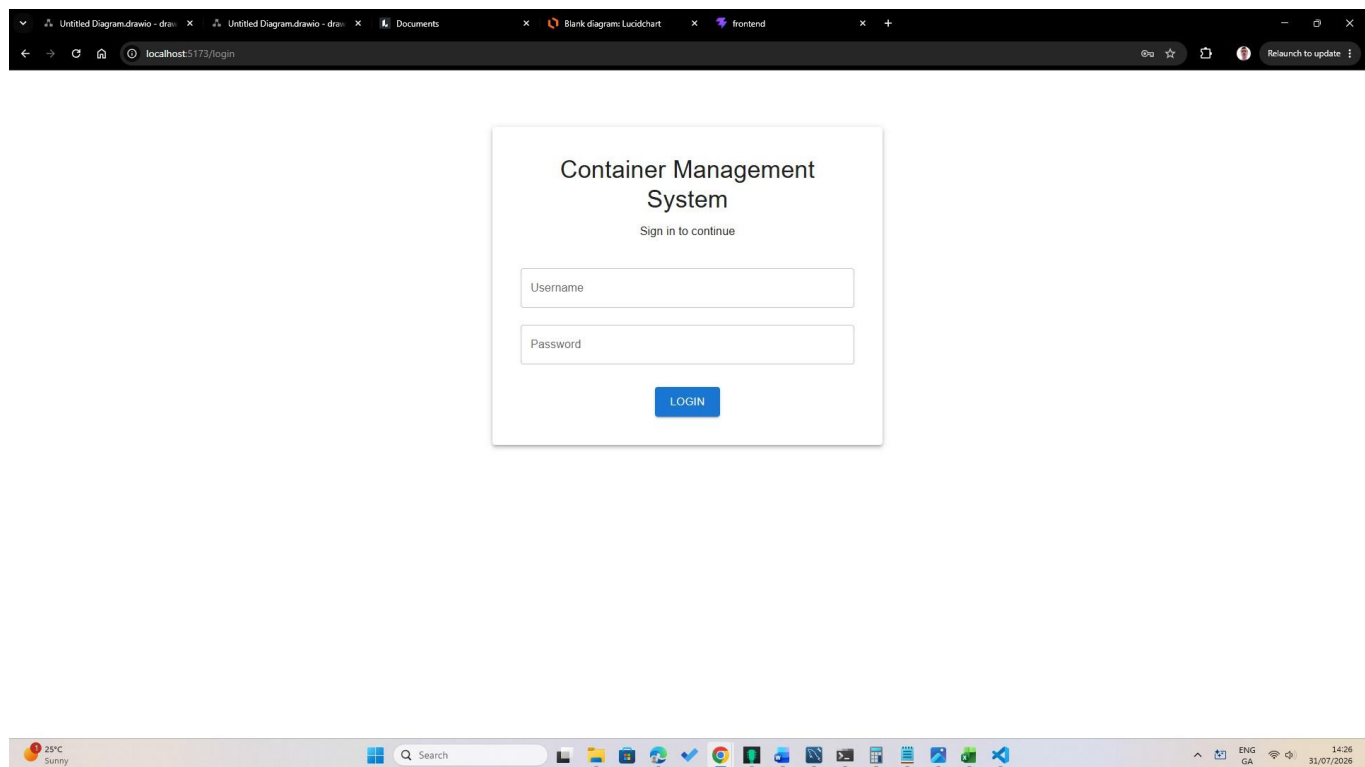


Figure 1: Login Screen

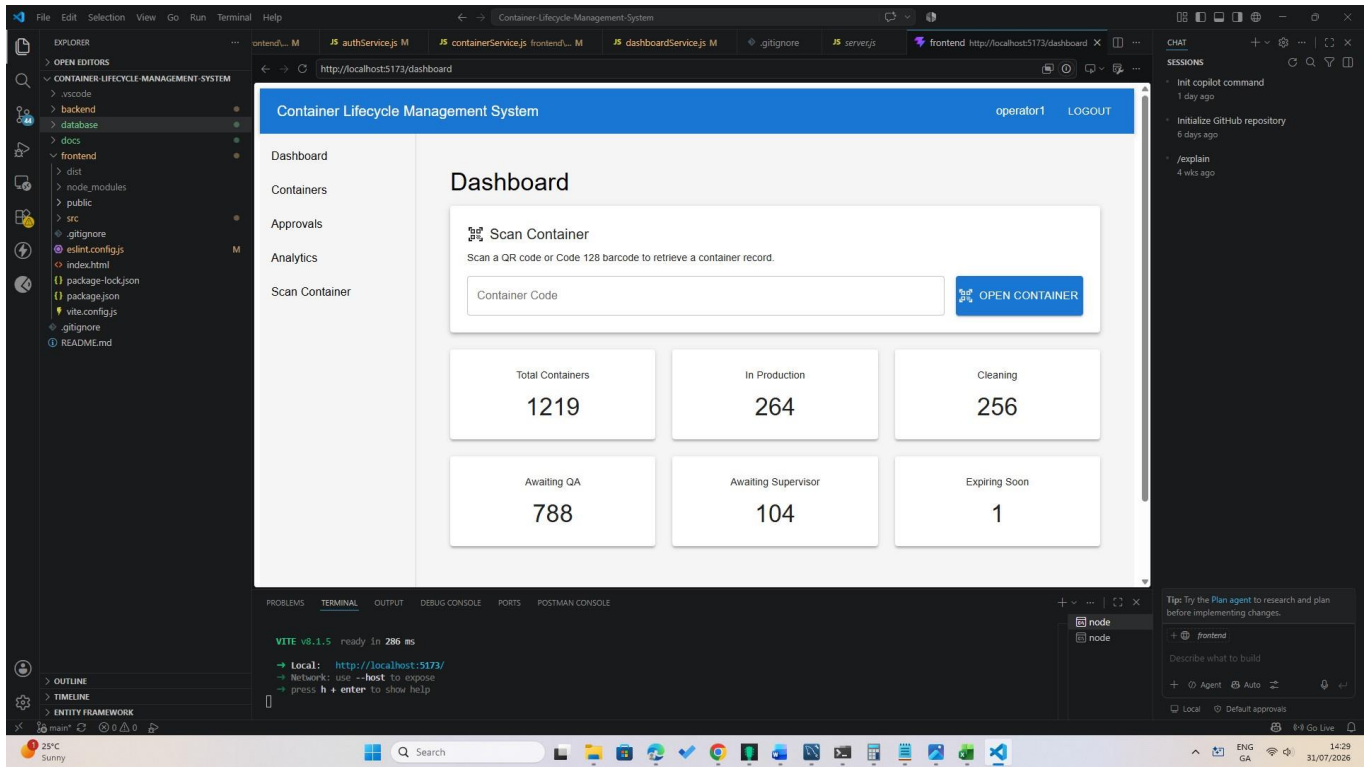


Figure 2. Dashboard

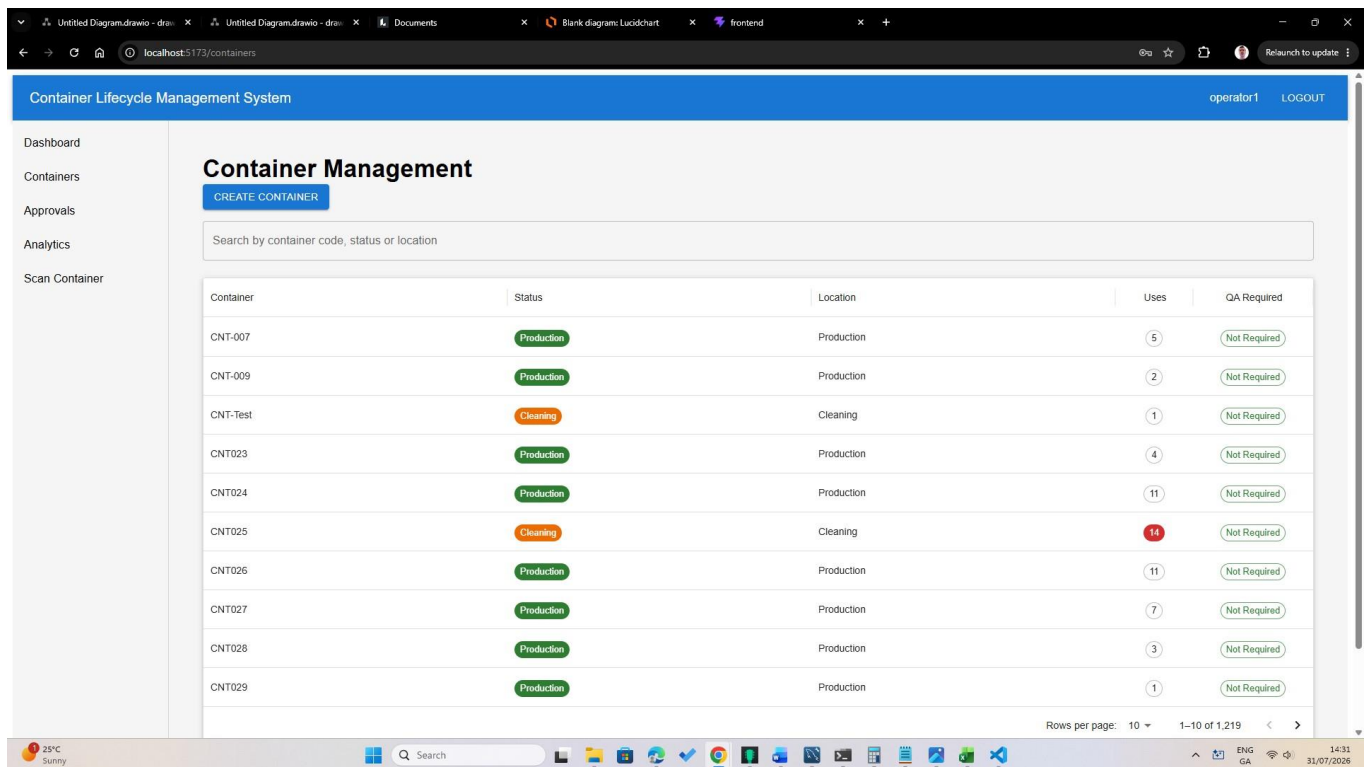


Figure 3. Container management showing the status and location of reusable production containers

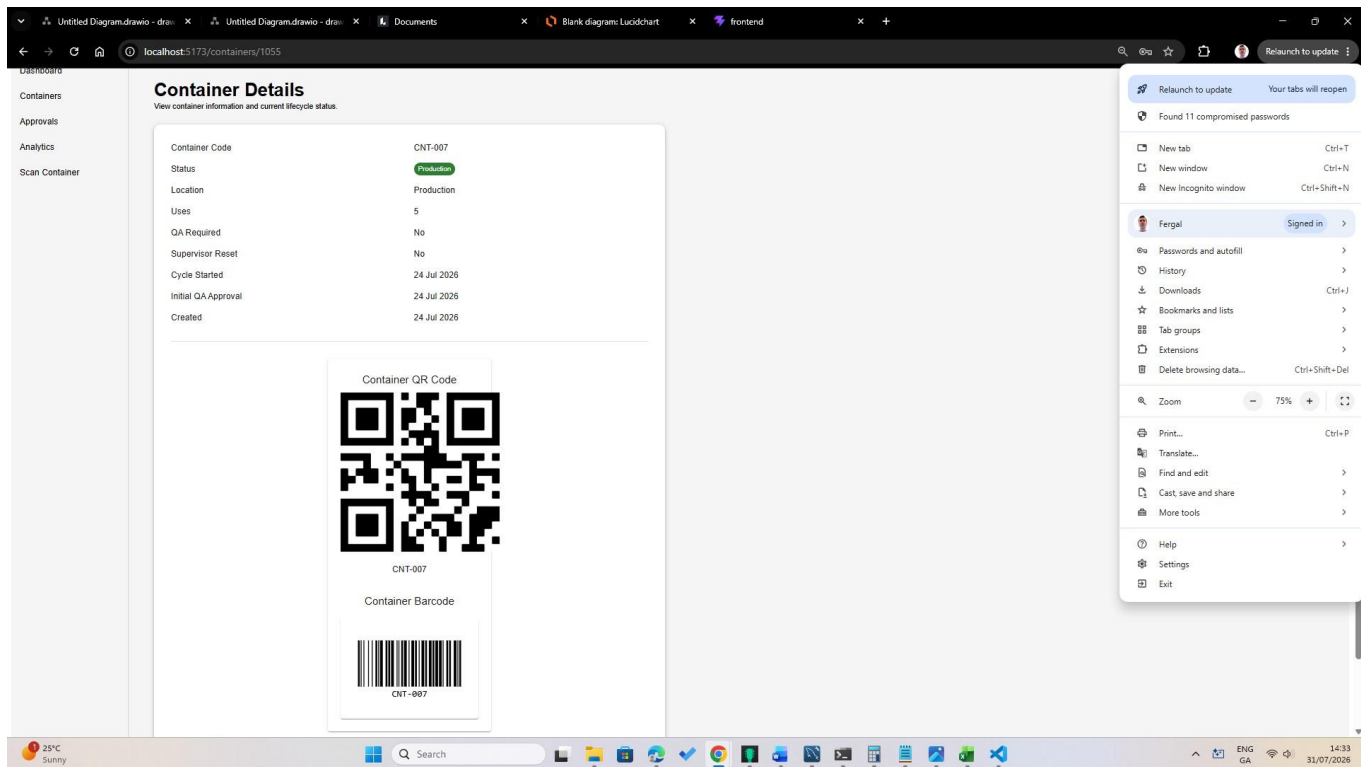


Figure 4. Container Details showing details and QR and Barcode

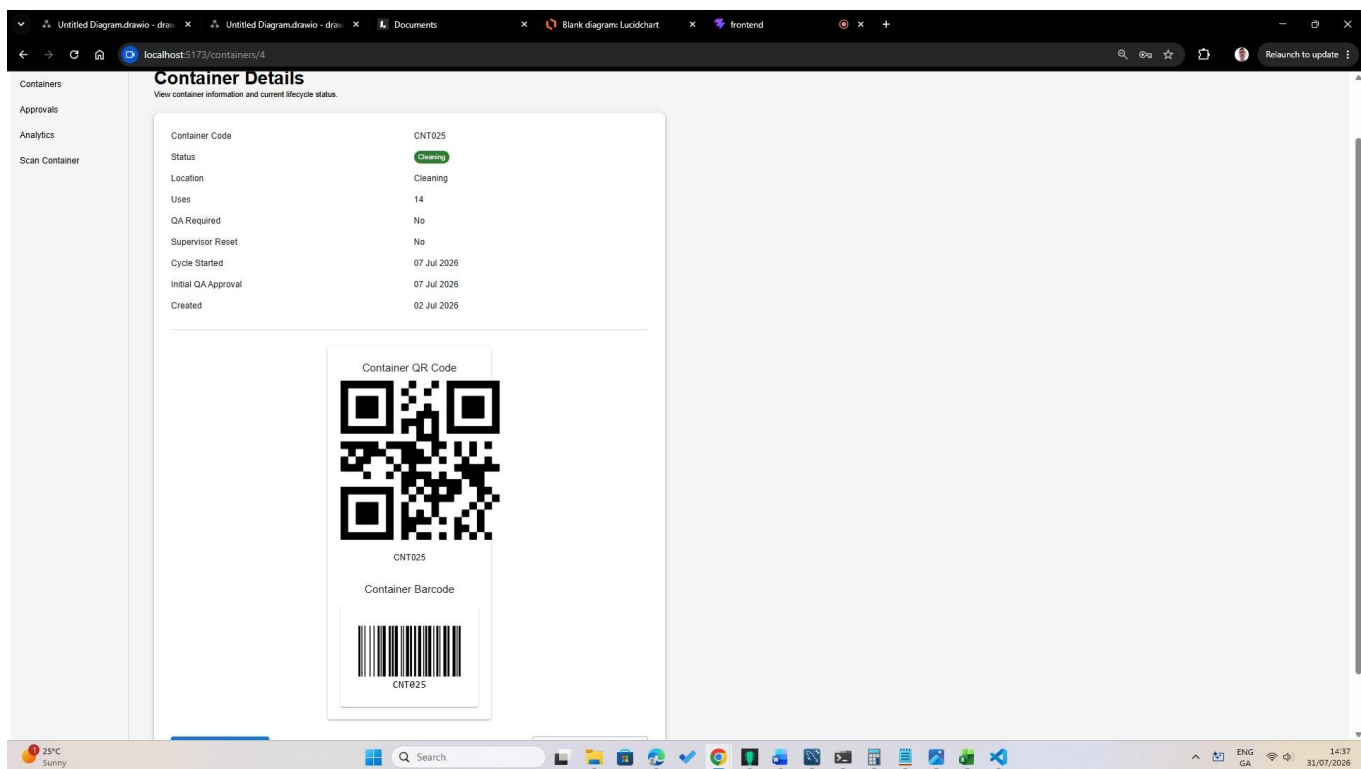


Figure 5. Container label QR was scanned and the details show on screen

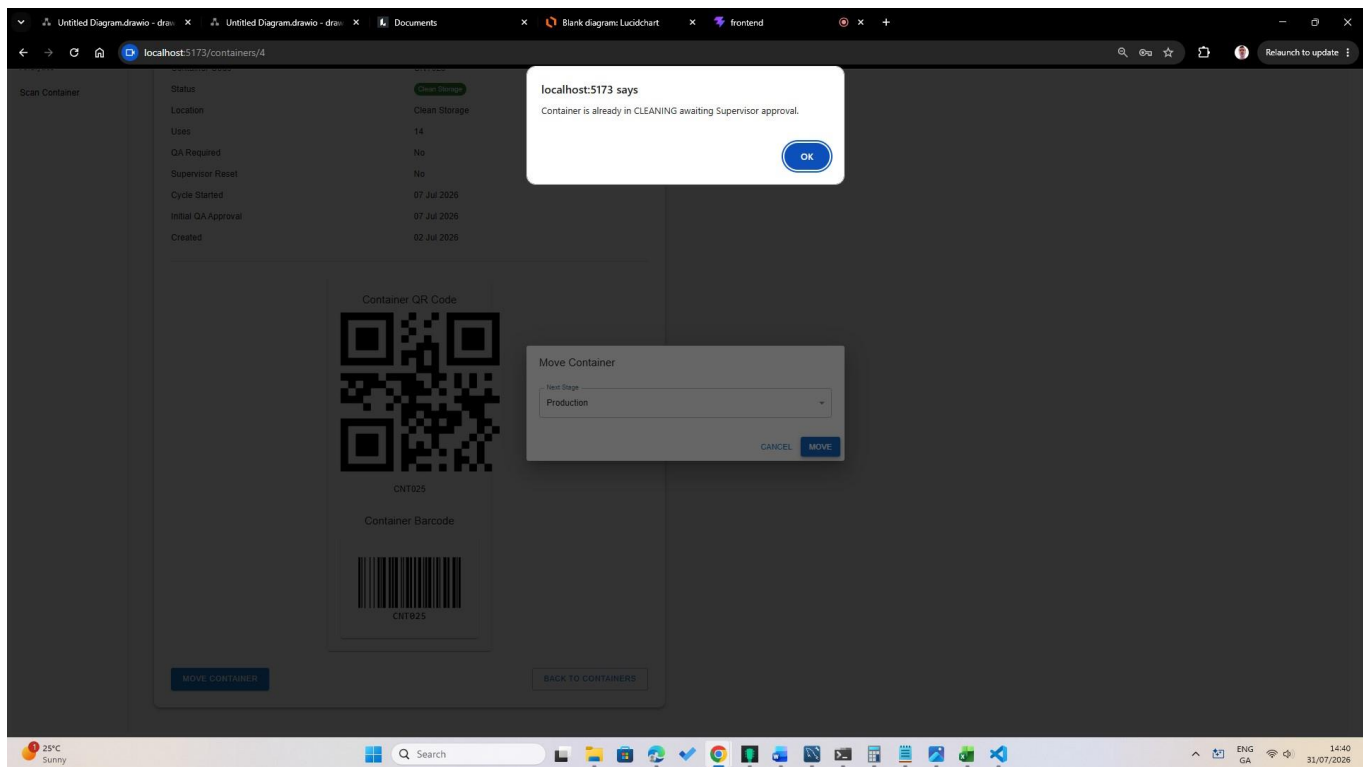


Figure 6. Container usage limit reached and alert showing before this container can be move to production it must be cleaned and approve by a supervisor.

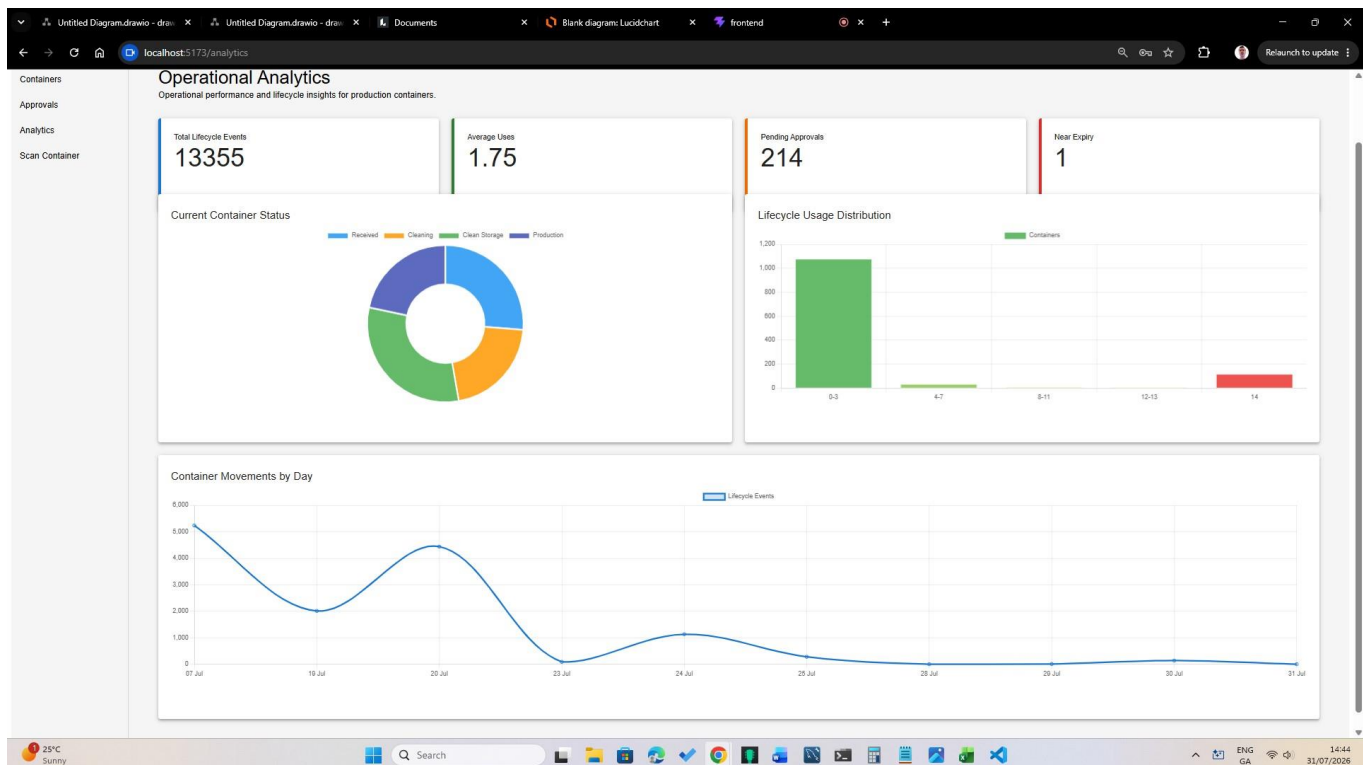


Figure 7. Container operational analytics to help improve decision making.

8. Project plan and Current Progress

The project has been managed using an Agile Scrum approach, with Trello used to organise tasks and monitor progress throughout development. Figma was used to design the user interface, while Draw.io and Lucid chart were used to create the system architecture, workflow, UML, and database diagrams.

Development has progressed in line with the project schedule. The database, backend API, authentication, workflow engine, React frontend, dashboards, QR code and barcode functionality, and simulation engine have been completed. The remaining work focuses on system testing, user interface refinements, operational data evaluation, and completion of the final dissertation.

Progress to date

Week	Planned	Current Status
Week 1	Requirements gathering, research and project planning	completed
Week 2	Database design and MySQL implementation	completed
Week 3	Backend API development testing	completed
Week 4	User authentication and workflow engine	completed
Week 5	Simulation engine for operational data	completed
Week 6	React Frontend development	completed
Week 7	Dashboard implementation and reporting	completed
Week 8	Operational analytics and data evaluation	In progress
Week 9	System testing, debugging and refinement	In progress
Week 10	Documentation and final project completion	In progress

9. Future Work

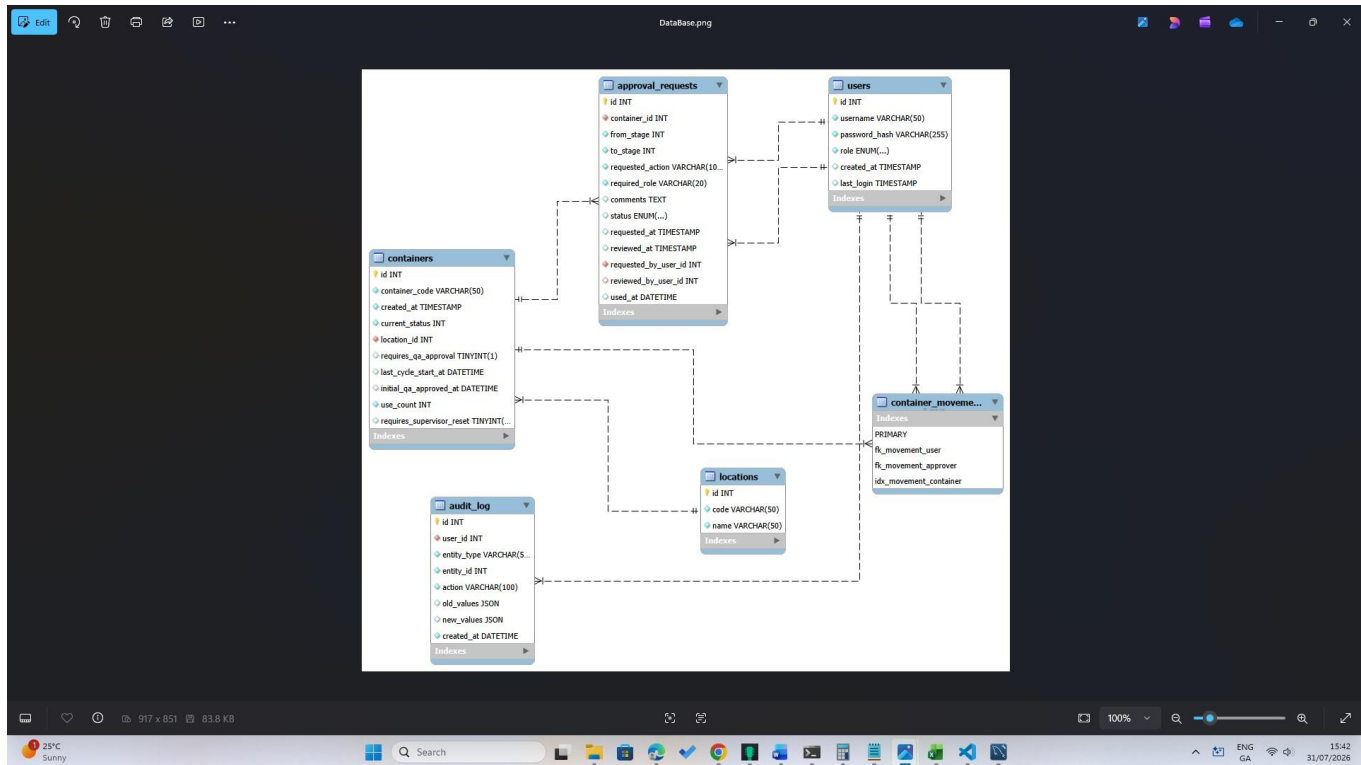
The remaining work on the project will focus on completing end-to-end system testing, refining the user interface, and evaluating the operational data generated by the simulation engine. Any issues identified during testing will be resolved to improve the reliability and usability of the application.

The final dissertation will also include a detailed evaluation of the completed system, supported by testing results and a discussion of how the project objectives were achieved. This will provide an assessment of the system's effectiveness in improving traceability, supporting regulatory compliance, and managing the lifecycle of reusable production containers.

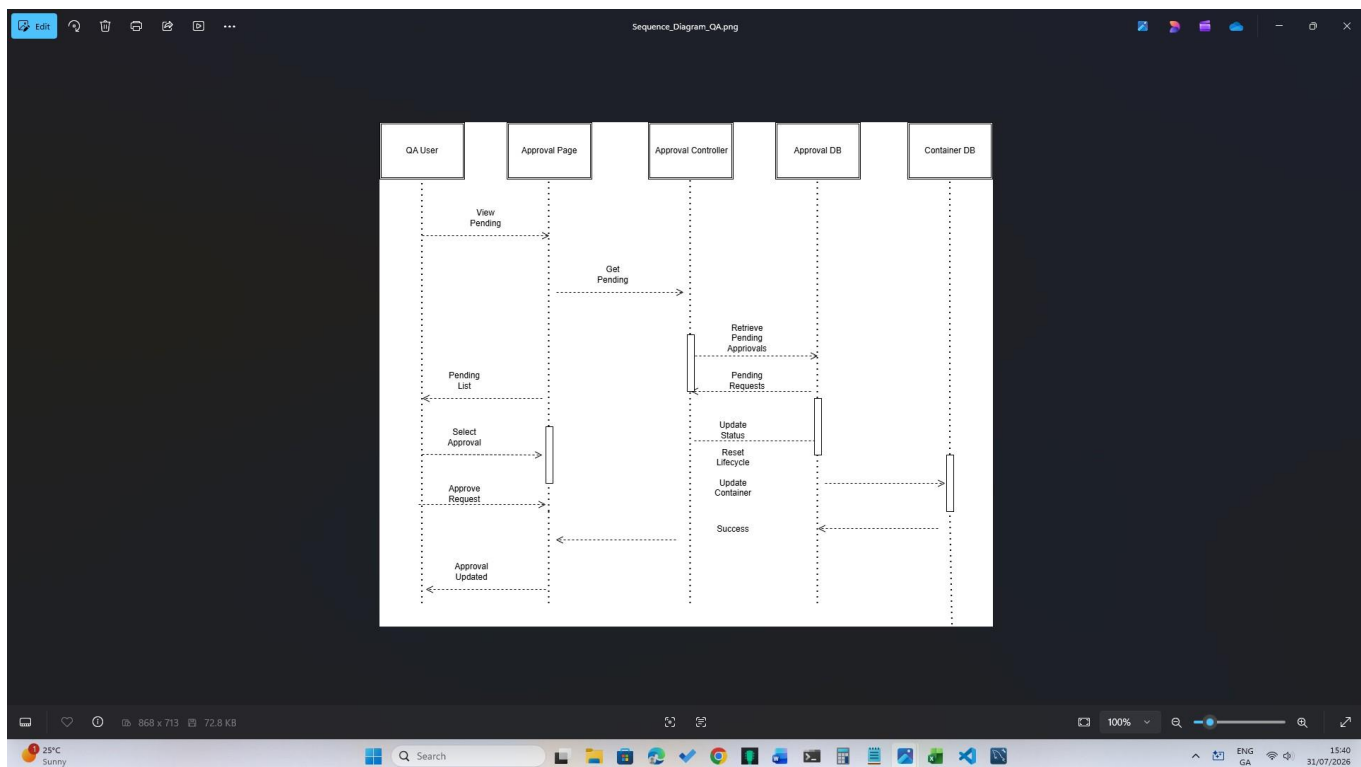
10. Conclusion

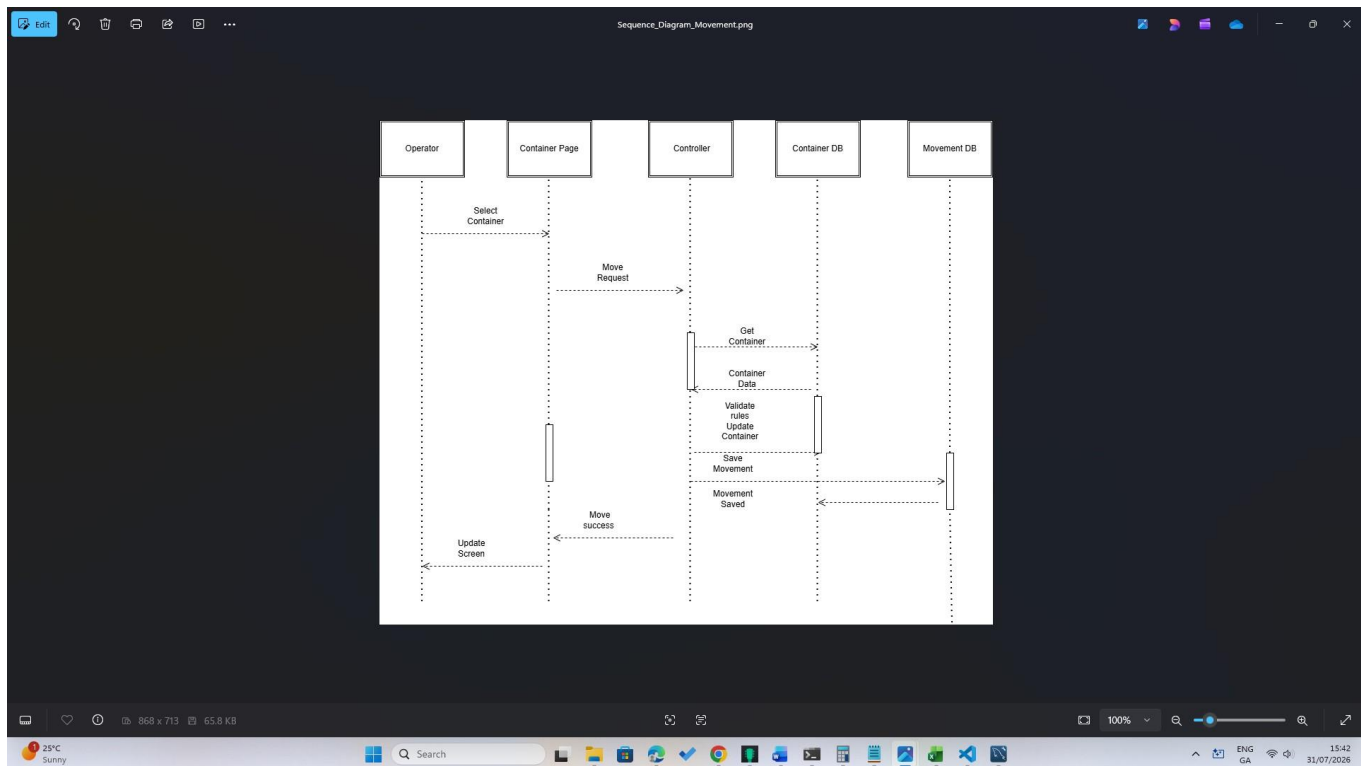
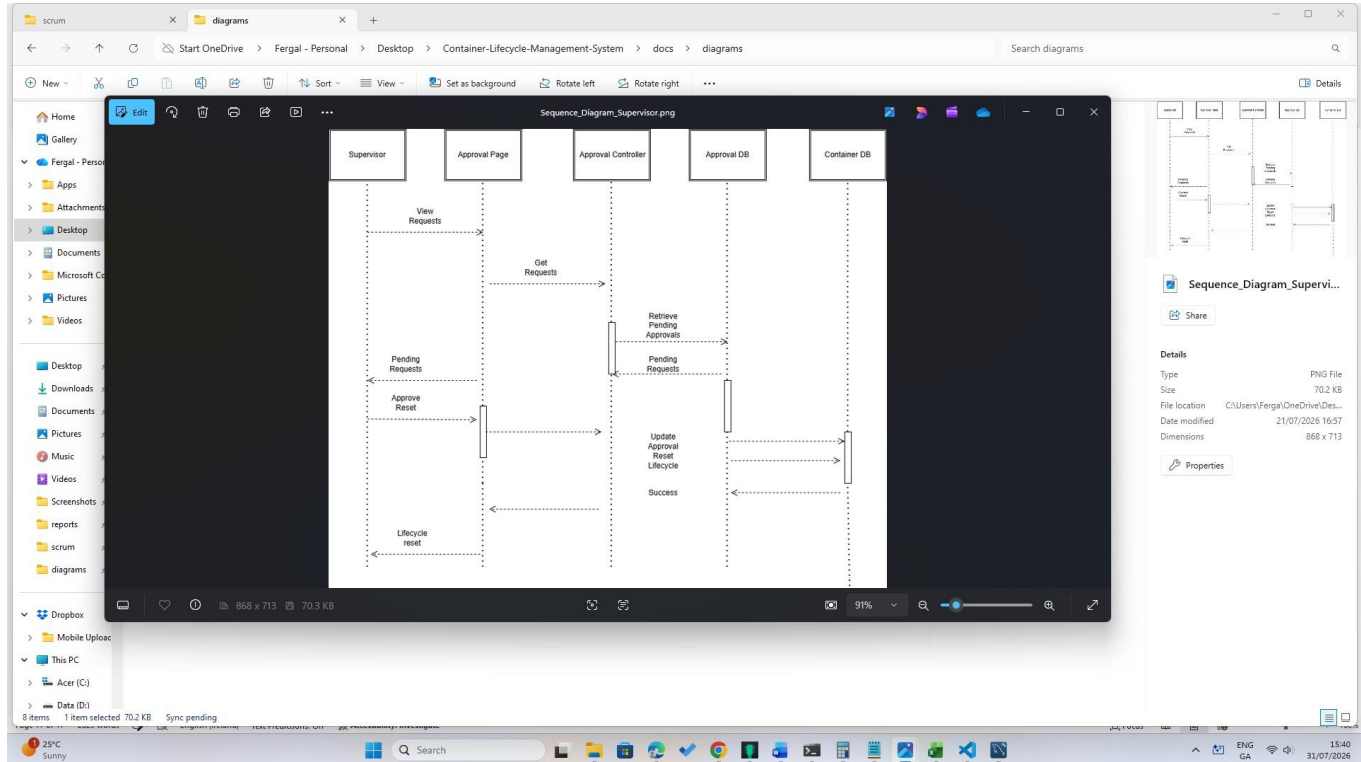
This project has made good progress towards developing a full-stack Container Management System for managing reusable production containers in a pharmaceutical manufacturing environment. The core features, including user authentication, container lifecycle management, approval workflows, movement tracking, dashboards, and QR code functionality, have been successfully implemented.

The remaining work will focus on completing system testing, evaluating the operational data generated by the simulation engine, refining the user interface, and preparing the final dissertation. Overall, the project remains on schedule and demonstrates a practical digital solution for improving traceability, workflow management, and regulatory compliance within a pharmaceutical manufacturing environment.



Appendix A - Entity Relationship Diagram (ERD) This illustrates the database schema and the relationships between the main entities used by the application.





Appendix B Container lifecycle movements and approvals that create movements in the application.